RESULTS REPORT

Model training, evaluation and findings

What was trained, how it was measured, and what it showed

DM-AHR-ViT · Explainable Vision Transformer for Skin Lesion Classification Using Dermoscopy Images

Research prototype only. Not clinically validated and not intended for diagnosis, triage or patient care.

In one sentence

Across 18 trained models (6 arms × 3 seeds) evaluated on a locked test set of 1,502 images, the best arm was A — original at macro-F1 0.750. Our first diffusion run (0.426) failed because of a training-script defect. After repairing it, retraining and teaching the model with real hair strands, the rebuilt arm Cv2 scored 0.720 ± 0.021, statistically indistinguishable from the original images, the 128 px control and classical DullRazor (section 8). On synthetic test pairs DullRazor recovered the clean image better than the diffusion arm. The pre-registered check that the rebuilt model removes real hair from photographs was not met, so this study does not show that hair removal helps or hurts classification, and removal on real photographs is supported only by a 12-image visual look and a pasted-strand test.

1. What was trained and how

Fifteen Vision Transformers were trained: five input conditions (arms) at three random seeds each. Everything except the arm is identical — same frozen split, same manifests, same architecture, hyperparameters, augmentation and evaluation code — so a difference between arms is attributable to the hair-removal condition alone. Creleased is the authors' released DM-AHR checkpoint run through our pipeline at a 25-step respaced schedule; it was added after the protocol was frozen and is reported as exploratory.

Arm	Input the classifier sees	Macro-F1 (mean ± SD)
A — original	untouched dermoscopy image	0.750 ± 0.005
B — DullRazor classical	classical hair removal (DullRazor-style)	0.698 ± 0.009
C — first diffusion run (defective trainer)	diffusion-generated hairless image	0.426 ± 0.013
A* — 128 px control	original through the same 128 px round-trip	0.732 ± 0.008
Creleased — released ckpt	released DM-AHR checkpoint, 25-step respaced (exploratory)	0.713 ± 0.028

Training recipe (identical for all fifteen runs)

Component	Setting
Backbone	ViT-B/16, ImageNet-1k pretrained, 224 px input, 196 patches
Optimiser	AdamW, lr 3e-5, weight decay 0.05, warm-up 5 epochs then cosine decay
Batch / precision	32, bf16 autocast, gradient clipping 1.0
Class imbalance	inverse-frequency class weights from the training split only
Early stopping	validation macro-F1, patience 10, max 100 epochs
Seeds	42, 123, 2026
Test protocol	the locked test set is scored once per frozen checkpoint

Compute environment

All GPU work ran on Modal: NVIDIA A100 80GB PCIe, Python 3.11.12, torch 2.5.1+cu124, CUDA 12.4. A typical training run took about 6.5 minutes (25 epochs to early stopping). Per-run GPU type and wall-clock are logged in artifacts/logs/run_registry.jsonl.

2. Classification results

All metrics are computed on the held-out test set of 1,502 images. Macro-F1 is the primary endpoint because classes are severely imbalanced: nv is two thirds of the data, so plain accuracy rewards ignoring the rare classes.

Arm	Macro-F1	Accuracy	Balanced acc.	Macro AUC
A — original	0.750 ± 0.005	0.858 ± 0.012	0.735	0.969

Arm	Macro-F1	Accuracy	Balanced acc.	Macro AUC
A* — 128 px control	0.732 ± 0.008	0.851 ± 0.006	0.726	0.969
B — DullRazor classical	0.698 ± 0.009	0.842 ± 0.005	0.687	0.965
C — first diffusion run (defective trainer)	0.426 ± 0.013	0.685 ± 0.019	0.452	0.878
Creleased — released ckpt	0.713 ± 0.028	0.848 ± 0.009	0.696	0.967

Hair-present subset

Arm	Macro-F1	n
A — original	0.749 ± 0.002	1316
A* — 128 px control	0.735 ± 0.013	1316
B — DullRazor classical	0.691 ± 0.009	1316
C — first diffusion run (defective trainer)	0.411 ± 0.017	1316
Creleased — released ckpt	0.712 ± 0.029	1316

This is the subset where hair removal actually operates — the images it could help. The ordering is unchanged from the full test set.

Per-class test metrics

Class	Arm	Support	F1	Recall (sensitivity)	Specificity	AUC
akiec	A	49	0.625	0.694	0.982	0.956
	B	49	0.589	0.639	0.982	0.947
	C	49	0.286	0.408	0.952	0.911
	Astar	49	0.556	0.673	0.974	0.961
	Creleased	49	0.562	0.633	0.979	0.954
bcc	A	77	0.750	0.706	0.991	0.975
	B	77	0.724	0.740	0.984	0.979
	C	77	0.315	0.290	0.971	0.881
	Astar	77	0.747	0.719	0.989	0.982
	Creleased	77	0.726	0.736	0.984	0.976
bkl	A	165	0.676	0.644	0.967	0.955
	B	165	0.628	0.590	0.965	0.947
	C	165	0.356	0.461	0.863	0.791
	Astar	165	0.677	0.630	0.971	0.949
	Creleased	165	0.692	0.679	0.965	0.946
df	A	17	0.733	0.706	0.998	0.987
	B	17	0.591	0.529	0.997	0.978
	C	17	0.140	0.137	0.990	0.863

Class	Arm	Support	F1	Recall (sensitivity)	Specificity	AUC
	Astar	17	0.719	0.627	0.999	0.985
	Creleased	17	0.620	0.471	1.000	0.980
mel	A	167	0.634	0.615	0.959	0.939
	B	167	0.561	0.529	0.957	0.935
	C	167	0.400	0.373	0.939	0.820
	Astar	167	0.613	0.611	0.952	0.925
	Creleased	167	0.609	0.617	0.949	0.936
nv	A	1006	0.943	0.956	0.855	0.981
	B	1006	0.942	0.958	0.844	0.980
	C	1006	0.871	0.827	0.855	0.922
	Astar	1006	0.943	0.949	0.872	0.982
	Creleased	1006	0.942	0.941	0.884	0.980
vasc	A	21	0.889	0.825	1.000	0.991
	B	21	0.853	0.825	0.998	0.991
	C	21	0.615	0.667	0.993	0.955
	Astar	21	0.868	0.873	0.998	0.999
	Creleased	21	0.840	0.794	0.999	0.994

Recall is the proposal's sensitivity. Supports for df (17) and vasc (21) are small, so their per-class numbers are noisy and are not interpreted individually.

3. Hair-removal image quality

Measured on the 186 held-out synthetic pairs — hair composited onto genuinely hair-absent images, so a true hairless reference exists. The unprocessed hairy input is scored as a floor: falling below it means the method made the image worse than doing nothing.

Condition	PSNR (dB) mean ± SD	SSIM mean ± SD
Hairy input (floor)	22.13 ± 1.90	0.520 ± 0.093
Arm B — classical	35.73 ± 2.11	0.918 ± 0.023
Creleased — released ckpt	27.08 ± 2.63	0.703 ± 0.094
Arm C — diffusion	14.29 ± 1.09	0.074 ± 0.006

Why Arm C scores below the untouched input — a training-script defect

An earlier version of this report called the checkpoint undertrained; that diagnosis is retracted. Our training script fed the network t/T as the noise level while the sampler feeds sqrt(alphas_cumprod_prev), and the two run in opposite directions. On 50 validation pairs the released checkpoint prefers the sampler's encoding at every timestep tested (2.9× to 20.5× lower error) and ours prefers t/T (about 10× lower), so more iterations of the old script could not have fixed it. Arm C as reported measures a defective training run and says nothing about DM-AHR as published. The script is now fixed; a retrain on validation pairs still failed full 2,000-step sampling (14.96 dB against a 21.30 dB hairy-input floor, 24 pairs) and was stopped on cost at 56,300 of a planned 100,000 iterations. See reports/oneshot_diagnostic_report.md.

Does any arm remove real hair? Not cleanly

Every image-quality number above is measured on synthetic hair overlays. A separate check on 40 hairy training images with real hair (test set not used; artifacts/diagnosis/real_hair_check.json) found that the released DM-AHR model leaves real hair in place: it changes hair pixels about as much as other pixels (ratio 1.13, against 21.7 for DullRazor), and the strands are visibly intact in the contact sheet. Our DullRazor arm does remove hair, but in all 8 images reviewed it replaced lesion and skin texture with blocky patches, and in one it erased the whole lesion; the likely cause is our frozen detector and inpainting settings. So neither arm cleanly removed real hair, and the classification comparison does not test whether hair removal helps. One control in that check failed (the hair-coverage measure, now void); the conclusion rests on the change ratio, whose control passed. See reports/hair_check_prereg.md.

4. Explainability

For every test image, two saliency methods — Grad-CAM on the patch grid and attention roll-out — were compared against the released HAM10000 lesion mask. The table reports the fraction of the top-20% most salient pixels that fall inside the mask, averaged over seeds, with the two chance baselines for context.

Arm	Grad-CAM overlap	Roll-out overlap	Grad-CAM pointing	Roll-out pointing	Uniform chance	Gaussian chance
A — original	0.363	0.555	0.468	0.842	0.200	0.644
A* — 128 px control	0.404	0.577	0.536	0.858	0.200	0.644
B — DullRazor classical	0.388	0.586	0.494	0.876	0.200	0.644
C — first diffusion run (defective trainer)	0.412	0.337	0.617	0.242	0.200	0.644
Creleased — released ckpt	0.382	0.568	0.524	0.878	0.200	0.644

Overlap measures spatial alignment with released masks, not clinically correct reasoning. Every arm beats uniform chance, but a centred-Gaussian prior scores 0.644 on overlap and 0.953 on the pointing game, exceeding every arm on both measures. No arm therefore beats a trivial attend-to-the-centre baseline, and these numbers do not establish that any arm localises lesions better than that prior. The masks themselves are single-rater: a dermatologist (Tschandl P) created binary masks for all 10,015 images, network-initialised and then verified, corrected or replaced by hand in FIJI (Tschandl et al., Nature Medicine 2020), so overlap is measured against one tracing, not consensus.

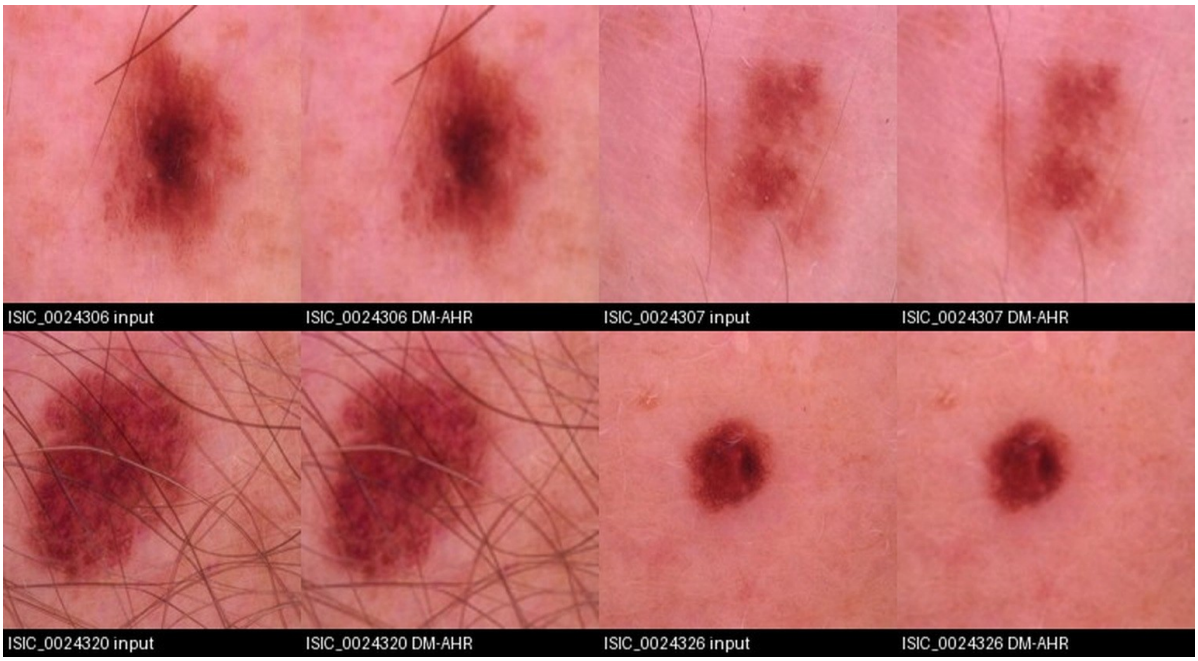


Figure 1 — Arm C diffusion removal, from the preprocessing report, reproduced here because the same failed hair-removal output drives both its quality scores and its classification deficit.

5. Statistical comparisons

Every endpoint is tested twice, as the protocol requires: a lesion-group bootstrap (10,000 replicates, the same resample shared across arms and seeds, percentile 95% CI) and a Wilcoxon signed-rank test on values collapsed to one per lesion group. For the classification endpoint the per-group value is that group's mean per-image correctness, so the Wilcoxon test asks whether the arms differ in per-lesion accuracy — it is not a test of macro-F1 itself, and the column header says so. Holm correction is applied across the three planned comparisons within each endpoint, separately for each test; A–A* is a secondary control in its own family and cannot inflate the planned family. A CI spanning zero is called inconclusive, not 'no difference'.

Endpoint	Comparison	Diff.	95% CI (bootstrap)	p (Holm)	Wilcoxon p (per-group acc.)	Groups
paired_psnr	B-hairy input	+13.604	+13.261 to +13.943	<0.0001	<0.0001	186
paired_psnr	C-B	-21.441	-21.768 to -21.106	<0.0001	<0.0001	186
paired_psnr	C-hairy input	-7.838	-8.088 to -7.592	<0.0001	<0.0001	186
paired_ssim	B-hairy input	+0.398	+0.385 to +0.410	<0.0001	<0.0001	186
paired_ssim	C-B	-0.843	-0.847 to -0.840	<0.0001	<0.0001	186
paired_ssim	C-hairy input	-0.446	-0.459 to -0.433	<0.0001	<0.0001	186
test_macro_f1	A-Astar	+0.018	-0.008 to +0.045	0.1668	0.0739	1437
test_macro_f1	A-B	+0.052	+0.019 to +0.087	0.0012	0.0113	1437
test_macro_f1	A-C	+0.324	+0.283 to +0.366	<0.0001	<0.0001	1437

Endpoint	Comparison	Diff.	95% CI (bootstrap)	p (Holm)	Wilcoxon p (per-group acc.)	Groups
test_macro_f1	B-C	+0.272	+0.225 to +0.320	<0.0001	<0.0001	1437
test_macro_f1	C-Astar	-0.306	-0.348 to -0.265	<0.0001	<0.0001	1437
test_macro_f1	A-Astar	+0.014	-0.014 to +0.041	0.3218	0.0800	1251
test_macro_f1	A-B	+0.058	+0.022 to +0.095	0.0008	0.0058	1251
test_macro_f1	A-C	+0.337	+0.291 to +0.383	<0.0001	<0.0001	1251
test_macro_f1	B-C	+0.279	+0.224 to +0.333	<0.0001	<0.0001	1251
test_macro_f1	C-Astar	-0.324	-0.370 to -0.279	<0.0001	<0.0001	1251
x1_gradcam_overlap20	A-B	-0.027	-0.039 to -0.017	<0.0001	<0.0001	1437
x1_gradcam_overlap20	A-C	-0.053	-0.065 to -0.040	<0.0001	<0.0001	1437
x1_gradcam_overlap20	B-C	-0.025	-0.037 to -0.013	0.0002	<0.0001	1437
x1_rollout_overlap20	A-B	-0.032	-0.034 to -0.029	<0.0001	<0.0001	1437
x1_rollout_overlap20	A-C	+0.217	+0.208 to +0.226	<0.0001	<0.0001	1437
x1_rollout_overlap20	B-C	+0.249	+0.239 to +0.258	<0.0001	<0.0001	1437

Exploratory: released-checkpoint arm

These rows are not part of the pre-registered family. Creleased uses the authors' released DM-AHR checkpoint at a 25-step respaced schedule, a deviation from the paper's 2,000 steps, and its classifier was trained on our budget. Holm correction is applied within this exploratory family only.

Endpoint	Comparison	Diff.	95% CI (bootstrap)	p (Holm)	Cohen's dz	Groups
paired_psnr	Creleased-B	-8.657	-9.063 to -8.254	<0.0001	-3.073	186
paired_psnr	Creleased-hairy input	+4.946	+4.503 to +5.388	<0.0001	+1.639	186
paired_ssim	Creleased-B	-0.215	-0.228 to -0.204	<0.0001	-2.558	186
paired_ssim	Creleased-hairy input	+0.182	+0.166 to +0.199	<0.0001	+1.595	186
test_macro_f1	A-Creleased	+0.038	+0.005 to +0.071	0.0476	+0.053	1437
test_macro_f1	B-Creleased	-0.015	-0.049 to +0.019	0.4004	-0.011	1437
test_macro_f1	A-Creleased	+0.037	+0.002 to +0.073	0.0768	+0.060	1251
test_macro_f1	B-Creleased	-0.021	-0.057 to +0.016	0.2614	-0.016	1251

6. What this says about the three hypotheses

H1 — image quality (diffusion preserves texture better than classical):

- PSNR: B higher than C (C - B = -21.441, 95% CI -21.768 to -21.106). SSIM: B higher than C (C - B = -0.843, 95% CI -0.847 to -0.840). H1 is not supported by these data. Arm C's SSIM is near-constant across all 186 pairs (0.074 ± 0.006, with 186 below 0.15), which is what the inverted noise-level signal in our training script produces. The released checkpoint was then scored through the same adapter (Creleased): it clears the hairy-input floor by 4.95 dB but sits 8.66 dB below classical inpainting, so H1 is now tested for DM-AHR and is not supported.

H2 — classification (diffusion-cleaned images improve the ViT):

- A vs C: A higher than C (A - C = +0.324, 95% CI +0.283 to +0.366). B vs C: B higher than C (B - C = +0.272, 95% CI +0.225 to +0.320). The direction is opposite to the hypothesis and the gap is large. The released checkpoint does not rescue it: A vs Creleased is +0.038 (95% CI +0.005 to +0.071, Holm p=0.048, negligible effect) and B vs Creleased is inconclusive (-0.015, CI -0.049 to +0.019, Holm p=0.400).
- A vs B: A higher than B (A - B = +0.052, 95% CI +0.019 to +0.087). This must not be read as evidence that hair removal hurts classification: the real-hair check shows our frozen DullRazor settings damage lesions, so the gap is confounded with image damage and was not separated from it.

H3 — explainability (attention focuses more on lesions after diffusion):

- Grad-CAM overlap, A vs C: C higher than A (A - C = -0.053, 95% CI -0.065 to -0.040). Attention roll-out, A vs C: A higher than C (A - C = +0.217, 95% CI +0.208 to +0.226). The methods disagree in direction, but H3 is not supported and the measurement is uninformative as specified: a centred-Gaussian prior beats every arm on both overlap (0.644) and pointing (0.953), so no arm demonstrates lesion localisation beyond a centre bias. That is a limitation of the measurement, not a finding about diffusion hair removal, and it must be fixed before H3 can be tested.

Resolution control:

- A vs A*: inconclusive (A - A* 95% CI -0.008 to +0.045). Because this is inconclusive, the 128 px round-trip alone does not explain Arm C's deficit.

7. Limitations

- **Arm C is a failed replication, caused by a training-script defect —** Our script gave the network an inverted noise-level signal (t/T instead of the sampler's sqrt(alphas_cumprod_prev)), so the model ran 20,000 iterations learning the wrong mapping; it was also sampled at 1,000 steps against 2,000 and upsampled from 128 px to 224. The upstream config's n_iter is 660,000 and the released checkpoint is named I725000_E1083. The fixed script, sampled from pure noise at the budgets tried, did not give usable output; the partial-noise sampler used for Cv2 did. Arm C's results describe that defective run, not the published method. The rebuilt arm Cv2 (section 8) is the fair test of the method.
- **Arm B damages lesions on real hair —** On 8 real-hair training images reviewed, our frozen DullRazor settings replaced lesion and skin texture with blocky inpainting patches, and in one image erased the lesion (a dark lesion and marker ink were read as hair). The A-versus-B difference is therefore confounded and is not evidence about hair removal in general. Re-tuning the arm on training and validation images would address it; that was not done.
- **The released diffusion model leaves real hair —** It reproduces synthetic hair removal well, which is what it was trained on, but on 40 real-hair training images it left the hair in place. Creleased is therefore close to 'the original at 128 px' and is not a test of hair removal.

- **Creleased is exploratory, not pre-registered —** It uses the authors' released checkpoint at a 25-step respaced schedule, a deviation from the paper's 2,000 steps, and the routine that produced those images is not reproducible from committed code. Its classifier was trained on our budget rather than the paper's. The paper states the model was trained on HAM10000 images other than a 700-image test set, so it very probably saw our validation and test images (no labels were used). Its comparisons are reported in their own Holm family and must not be read as pre-registered results.
- **Three seeds —** inference is over ~1,437 lesion groups with seeds averaged inside each replicate, never counted as independent samples; still, three seeds cannot characterise variance exhaustively.
- **Screening unvalidated —** the proposal specifies the manually annotated hair masks of Lama et al. (ChimeraNet) as an independent reference for screening quality, and those masks were not obtained. This is a skipped step, not a gap in the data. The threshold was calibrated on synthetic pairs and frozen, which protects the test set but does not validate the detector on real hair. The 89.8% screening prevalence is reported as a limitation.
- **Masks are single-rater —** the HAM10000 segmentations were created by one dermatologist (Tschandl P), network-initialised and then verified, corrected or replaced by hand in FIJI (Tschandl et al., Nature Medicine 2020). No inter-rater agreement is available, so explanation overlap is measured against one tracing.
- **Single corpus —** one public dataset; nothing here establishes behaviour on other scanners or populations.
- **Licensing unresolved —** HAM10000 is CC BY-NC 4.0; the DM-AHR repository's README (CC BY-NC 4.0) and LICENSE (Apache-2.0) contradict each other. Commercial use needs a rights decision.

8. Stage C: the rebuilt diffusion arm (Cv2)

The first diffusion run (arm C) failed because of the training defect described above. The trainer was repaired and the model retrained. A partial-noise sampler (start from the noised input at level 600 of 2,000, 50 strided DDIM steps) replaced sampling from pure noise, and the model was fine-tuned for 20,000 iterations on real hair strands harvested from train-split photos and pasted onto clean train-split photos. The procedure, checkpoint hash, sampler and endpoints were written down before any test-split use of the new model (reports/protocol_addendum_stageC.md); each of the three Cv2 classifiers was evaluated once on the test split.

Classification

Arm	Full test (n=1,502)	Hair-present (n=1,316)
A — original	0.750 ± 0.005	0.749 ± 0.002
A* — 128 px control	0.732 ± 0.008	0.735 ± 0.013
B — DullRazor classical	0.698 ± 0.009	0.691 ± 0.009
Cv2 — rebuilt diffusion	0.720 ± 0.021	0.714 ± 0.021
C — first diffusion run (defective trainer)	0.426 ± 0.013	0.411 ± 0.017

- **A vs B:** A higher than B (A - B = +0.052, 95% CI +0.019 to +0.087); Holm p 0.004.
- **A vs Cv2:** inconclusive (A - Cv2 95% CI -0.006 to +0.066); Holm p 0.198.
- **B vs Cv2:** inconclusive (B - Cv2 95% CI -0.052 to +0.009); Holm p 0.198.
- **Cv2 vs A* (resolution control):** inconclusive (Cv2 - A* 95% CI -0.040 to +0.016); Holm p 0.386.

Only the original-over-DullRazor difference is clear. The rebuilt arm cannot be told apart from the originals, the 128 px control or DullRazor. The secondary Wilcoxon test gives small p-values

for B vs Cv2 and Cv2 vs A*, but with effect sizes below 0.06 and bootstrap intervals that include zero these are not read as findings.

Image quality on the 186 synthetic test pairs

Input	PSNR (dB)	SSIM
Hairy input	22.13	0.520
B — DullRazor	35.73	0.918
Cv2 — rebuilt diffusion	29.18	0.861
Creleased (its own sampler)	27.08	0.703

Cv2 improves on the hairy input by +7.05 dB (95% CI +6.89 to +7.21) but is -6.55 dB against DullRazor (CI -6.91 to -6.19). The fidelity hypothesis is not supported on these pairs. The synthetic hair is simple dark curves that the detector finds exactly, DullRazor repairs only those pixels, and the diffusion model regenerates the whole 128 px image and loses fine detail. The pairs say nothing about real hair.

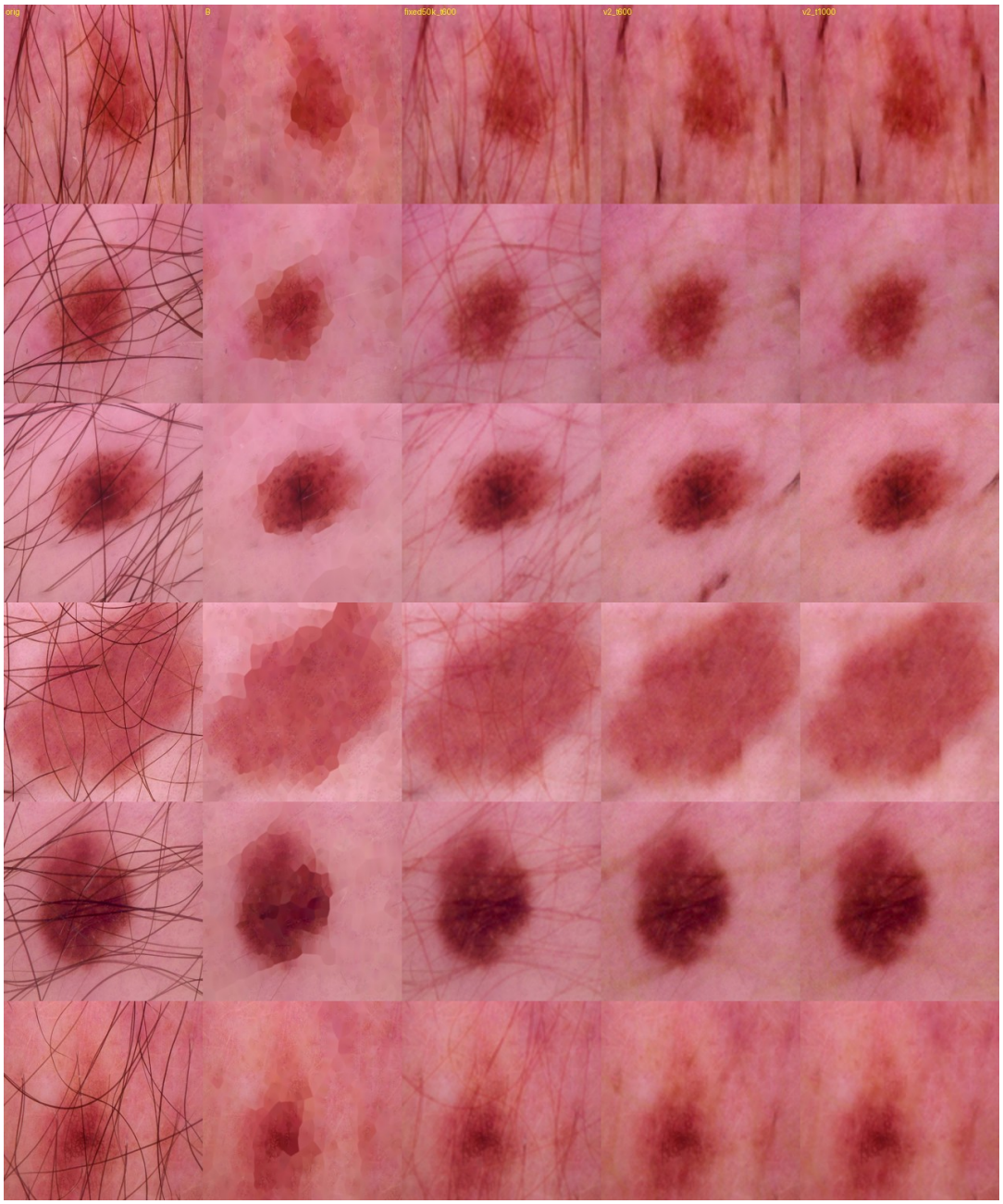
Explanation overlap (X1 subset; top 20% attention inside the lesion)

Arm	Grad-CAM overlap	Rollout overlap
A — original	0.364	0.558
A* — 128 px control	0.406	0.580
B — DullRazor classical	0.391	0.589
Cv2 — rebuilt diffusion	0.410	0.604

Against the resolution-matched control A*, Grad-CAM overlap does not differ (+0.004, CI -0.006 to +0.014), so the gain over A comes from the 128 px resolution. Rollout overlap is higher by +0.024 (CI +0.022 to +0.026), a small gain whose interval leaves out seed-to-seed variation; Cv2 exceeds A* in two of three seeds. The A* rows were added to the statistics after the planned rows had been seen.

What is and is not validated

- **Gate not met —** the Stage B gate for removing hair from real photographs was not met (inside/outside ratio 1.95 and 1.57 against 2.0; follow-up composite rule 0.14 and 0.15 against 0.20). Stage C ran anyway as a recorded deviation (decision D34).
- **What supports removal on photographs —** an agent-reviewed look at 12 dense-hair validation photos shows nearly all hair removed with lesion outline and colour kept and fine texture lost; the ground-truth test on pasted real strands removes 77% of strand error at t0=600. This is supporting evidence, not a passed pre-registered test.



Validation photos with dense hair. Columns: original, DullRazor, first fixed model, rebuilt model at t0=600, rebuilt model at t0=1000 (128 px, agent-reviewed).

9. Reproducibility

- **Determinism —** Arm A seed 42 re-run from scratch reproduced validation macro-F1 exactly (0.778336 vs 0.778336).
- **Portability —** the frozen split hash was reproduced byte for byte across two Modal accounts, and the full data-preparation chain was re-run natively with no Modal present, giving the identical hash.
- **How to explore —** notebooks/01_results_walkthrough.ipynb reads the committed artifacts and explains these tables interactively; notebooks/02 and 03 reproduce the pipeline on Modal or a local GPU.

Every table here is generated from artifacts/logs/: classification_seed_summary.csv, classification_per_class.csv, paired_psnr_ssim_test.csv, xai_localization_*.csv and statistical_comparisons.csv. This document is built by scripts/14_results_report.py and can be regenerated after any rerun.